

BRAC University
Department of Computer Science and Engineering
MSc/M.Engg. in CSE Program
CSE 710: Advanced Artificial Intelligence
Semester: Spring 2026

Faculty Information:

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Class:

CSE710 (Sec01): Saturday from 6:30 pm to 9:10 pm [Bangladesh Time]

Course Description:

Introduction, Advanced search techniques in AI; Planning and Knowledge representation; Uncertain Knowledge and Reasoning, Probabilistic Reasoning over time; Making simple and complex decisions; Knowledge in Learning, Learning Probabilistic Models, Graphical Models.

Pre-requisites: None

Course contents:

Fundamentals of AI, Genetic Algorithm, Knowledge Representation and Semantic Net, Graphical Models and D-separation, Probability theory and Bayes Net, Markov Decision Process, Expectation Maximization, Hidden Markov Model, Decision theory, Matching Game based Decision theory, Multi-Attribute Decision Making (MADM)-TOPSIS and Fuzzy-TOPSIS, Recommender Engine: Collaborative Filtering, Reinforcement Learning.

Course Outcomes:

[CO1] **Formulate** Artificial Intelligence (AI) problems for automated and intelligent decision-making.

[CO2] **Analyze** different AI approaches with their strengths and weaknesses and **select** appropriate AI algorithms based on the nature of the problem.

[CO3] **Implement** AI algorithms for solving real-world problems and write reports with analytical results.

Course Assessment Methods:

Homework

Homework/assignments shall be designed to ensure that the students have the required knowledge to analyze and design control systems. Specifically, they will support the student's progress in the project/Laboratory

Quizzes

Quizzes will be designed to test the student's understanding of the course and to assess various course outcomes

Examinations

The exam shall contain problems designed to test knowledge and comprehension, analyze control systems, and/or apply the engineering problem-solving method.

Textbooks:

- 1 Artificial Intelligence: A Modern Approach by Stuart Russel, Peter Norvig, 4th edition, 978-0-13-604259-4.
- 2 Machine Learning, Tom Mitchell.
- 3 Convex Optimization by S Boyd, L Vandenberghe. Cambridge University Press, 2004
- 4 Decision Support Systems and Intelligent Systems 8th edition, Turban & Aronson.
- 5 Multiple Criteria Decision Making: Methods, Examples and Python Implementations by Jason Papathanasiou and Nikolaos Ploskas (Springer, 2018)
- 6 Decision Making in the Manufacturing Environment using Graph Theory and Fuzzy Multiple Attribute Decision Making Methods, R. Venkata Rao, Vol-2, Springer, 2013.

Supporting Tools:

buX enrollment link:

Google Classroom Code and link: **rlphcekh**

Discord channel link:

Course repository:

https://drive.google.com/drive/folders/1B90wgRUzRU6uvYXgQRcPEdtpMPSr_7SB?usp=sharing

Course Policies:

Class Policy

- Classroom attendance is not mandatory. You should follow both the Google Classroom and bux contents and instructions.
- You should pay attention to the instructor and participate in class discussions.

Honor Code

Any form of cheating, plagiarism, and/or academic dishonesty will result in an "F" grade in the course.

Late Work and Examinations

Late assignments will not be accepted. Students who know that they are going to miss class should make arrangements in advance. Exams will be closed book. There will not be any make-up for quizzes and midterm exams except the cases of hospitalization or detention

Grading Policies:

Student's grades are assigned according to the grading scale of the Brac University Undergraduate Study and Examinations Regulations. In addition, the faculty are allowed to take into consideration the class average and standard deviation to reflect the actual class performance for student grade assignments. The grades at the university will be indicated in the following manner:

Marks	Grades
90-100	A (4.0)
85- <90	A- (3.7)
80- <85	B+ (3.3)
75- <80	B (3.0)
70- <75	B- (2.7)
65- <70	C+ (2.3)
60- <65	C (2.0)
57- <60	C- (1.7)
55- <57	D+ (1.3)
52- <55	D (1.0)
50- <52	D- (0.7)
<50	F (0.0)
P	Pass
I	Incomplete
W	Withdrawal
R	Retaken

Course Assessment Methods:

Guidelines for CSE course teaching in BRAC University. The following assessment methods are based on a theory course.

Assessment methods	Marks (%)
1. Attendance	05 %
2. Quiz	10%
3. Assignments & Presentations	15 %
4. Mid-Term Examination	25 %
5. Final Examinations	25%
6. Project Report/paper	20 %
Total	100 %

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